

Programme	B. Sc. STATISTICS				
Course Code	STA7 CJ 403 (P)				
Course Title	DISTRIBUTION THOERY				
Type of Course	Major				
Semester	VII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practicaum per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Basic knowledge of various univariate and bivariate distributions. 2. Matrix theory – Eigen Values & Eigen vectors.				
Course Summary	The main objective of this course are to understand the concepts of multivariate probability distributions. Study essential properties of multivariate distributions and apply customized probability distributions in the relevant context.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
1.	Distinguish different distributions and illustrate their role in modelling count data.	U	F	Seminar Presentation / Group Tutorial Work
2.	Describe the properties and applications of multivariate normal distribution.	U	C	Instructor-created exams
3.	Estimate the ML Estimates of the mean vector and dispersion matrix of multivariate normal.	An	P	Instructor-created exams
4.	Evaluate marginal and conditional distribution from multivariate normal distribution	An	P	Instructor-created exams / Home Assignments
5.	Describe the genesis of Wishart distribution with its properties.	U	C	Home Assignment
6.	Explain distribution function of random vectors, order statistics and their distributions.	Ap	C	Instructor-created exams / Home Assignments
7.	Compare Hotelling T^2 and	U	F	Instructor-created

	Mahalanobis D^2 statistic and able to apply them in testing problems.			exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45+30)	Marks (70)
I	Open-End		30	20
	1	Overview of univariate & bivariate distributions and their properties. Sampling Distributions- Central and non-central (t, F and χ^2) Order statistics – their distributions and properties- Joint, marginal and conditional distribution of order statistics The distribution of sample range and sample median. Problems using R/Python		
II	Multivariate Normal Distribution		12	20
	1	Definition and properties of multivariate normal density function		
	2	Distribution of a linear combination of the components of a normal random vector.		
	3	Maximum Likelihood estimation of the mean vector and dispersion matrix.		
	4	The distribution of sample mean vector inference concerning the mean vector when the dispersion matrix is known for single and two populations.		
III	Generalized Variance		9	15
	5	Wishart Distribution		
	6	Properties of Wishart distribution		
	7	Test for covariance matrix		
	8	Test for equality of covariance matrices		
	9	Test for independence of sets of variables.		
IV	Quadratic forms and their distributions		14	15
	10	Jacobian of matrix transformation of $Y=AXB$; $Y=AXA'$; $X=TT'$		
	11	Independence of a linear form and quadratic form		
	12	Distributions of quadratic form of a multivariate vector		
	13	Cochran's theorem		
	14	Partial and multiple correlation coefficients		
	15	Partial regression coefficients		
V	T^2 and D^2 distributions		10	
	16	Hotelling T^2 distribution and its applications		
	17	Generalized T^2 statistic and its distribution		
	18	Uses of T^2 statistic		
	19	Optimum properties of T^2 statistic		
	20	Mahalanobis D^2 statistic and its distribution		
	21	Relation between T^2 and D^2		
	22	Test based on T^2 statistic		

Text Book

1. **Anderson T W (2010)** : An Introduction to Multivariate Statistical Analysis, Wiley Eastern Ltd.
2. **Johnson, R A and Wichern D W (2003)** : Applied Multivariate Statistical Analysis, Prentice-Hall of India Private Ltd., New Delhi.

Reference

1. **Jhonson, Kotz and Balakrishna (1991)** : Continuous univariate distributions, Vol-1 2nd Ed., John Wiley and Sons
2. **Johnson, Kemp and Kotz (1992)** : Univariate Discrete distributions, 2nd Ed, John Wiley and Sons
3. **Kotz, Balakrishnan, Johnson (2004)** : Continuous Multivariate Distributions, Vol 1, 2nd Ed. John Wiley & Sons
4. **Mukhopadhyay P (1996)** : Mathematical Statistics, New Central Book Agency (P) Ltd. Calcutta.
5. **Srivastava, M, C G Khatri (1979)** : Introduction to Multivariate Statistics, Elsevier Science Ltd.